

Adam Torres Encarnacion

University Park, PA | [Email](#) ✉ | [LinkedIn](#)  | [GitHub](#)  | [Portfolio](#) ▲

EDUCATION

The Pennsylvania State University

B.S. in Data Science, B.S. in Statistics

Expected: Dec 2026

GPA: 3.7

WORK EXPERIENCE

Amazon

May 2026 – Aug 2026

Software Engineering Intern

Cupertino, CA

- Refactored C++ infrastructure for 2 fiber-cleaning robot arms projected to save \$28.6M/year across 85 data centers
- Built a 19-route HTTP server to control robot arms remotely over SSH while keeping the local control loop under 2ms
- Made an LLM-as-judge classification pipeline for 6 optical fiber and transceiver labels, storing live detection data in S3
- Created a self-evolving MCP knowledge base for agent-first development, migrating ~400 legacy docs from 2 repos
- Hardened 2 AWS CI/CD pipeline stages with blocking lint checks, code complexity checks, and ~40 integration tests

IBM

May 2025 – May 2026

AI Engineering Intern

Remote

- Shipped an LLM-powered REST microservice to power documentation for 3,100+ scripts, saving \$223,000 in labor costs
- Helped triage ~500k criminal records spanning 20+ years with a LangChain chatbot built on Python, cutting review time
- Deployed serverless Azure Function Apps that autoscaled with incoming jobs, cutting manual review workload by 78%
- Refined a file-chunking pipeline using Llama-4 and GPT-4o-mini to handle 7+ large-scale internal processing requests

PROJECTS

 **YADL (OpenAI WebMCP Challenge '26)** | *Python, OpenCV, MPS, PostgreSQL, AWS, Vercel, Railway*

- Turned AI agents into click-free CV data annotators through 26 WebMCP tools to draw & navigate an open canvas
- Built an OpenCV vision keyboard emulator with 98.78% F1 on 1,253 AI-labeled images after 600 MPS-accelerated epochs
- Balanced image datasets using 8 deterministic image transforms and GPT-Image-2 in asynchronous augmentation jobs
- Supported 16 concurrent labeling sessions at <300 ms p95 metadata API latency using FastAPI, PostgreSQL, and S3

 **nexdraw (NexHacks '26 ByteDance 2nd place winner)** | *TypeScript, Next.js, Gemini, PostgreSQL*

- Won 2nd (out of 1500+) solo with a tldraw canvas that auto-completes sketches in place via Gemini AI image generation
- Used Gemini to classify chat vs drawing requests, then cropped subjects with Grounding DINO on CUDA (PyTorch)
- Integrated OpenAI Whisper for voice prompts and gated each image generation behind an on-canvas accept/reject overlay
- Saved tldraw snapshots as PostgreSQL JSONB on a 2s debounce, generating PNG previews and skipping saves offline

 **simulacra (YHack '26 K2 Think V2 1st place winner)** | *Python, FastAPI, LangGraph, React*

- Built a multi-agent simulation engine with FastAPI and LangGraph StateGraph, using asyncio for concurrent agent execution, shared memory streams, and Pydantic validation for structured outputs
- Modeled echo-chamber formation across 50+ agents using recency-weighted memory retrieval, threshold-triggered reflection, and Deffuant bounded-confidence opinion updates; streamed simulation updates over Socket.IO

LEADERSHIP EXPERIENCE

Claude Builder Club

Aug 2025 – Jan 2026

Vice President

University Park, PA

- Brought Claude AI to Penn State, drawing in 260+ club signups in its first month (the fastest growth in school history)
- Taught agent harnesses, skills, hooks, MCP, and multiplexers to 170+ students, with 90% returning for follow-up sessions

Nittany AI

Jan 2025 – Jan 2026

Director (AI)

University Park, PA

- Led an AI research reading group with 40+ students per semester and 85% retention for the largest AI club on campus
- Covered ML papers [YOLO](#), [SAM3](#), [Attention Is All You Need](#), and [Adam Optimizer](#) over 5 months of student-led seminars

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, SQL, TypeScript, HTML/CSS, R, Lua

Frameworks: PyTorch, OpenCV, Hugging Face, Pandas, NumPy, FastAPI, React, Node.js, Next.js, LangChain/LangGraph

Developer Tools: Git, Linux, VS Code, Vim/Neovim, Pi (harness), Cmux (terminal multiplexer), Agile/Scrum

Databases: PostgreSQL, MySQL, SQLite, Redis, MongoDB, DynamoDB

Cloud/Infra: AWS (Lambda, S3, EC2, Bedrock, DynamoDB), Azure Functions, Vercel, Docker, CI/CD (GitHub Actions)